

Software Requirements and Design Document

For

Group 1

Version 1.0

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1. Overview (5 points)

The Gym Logger App is a web-based workout tracking system designed to give users a centralized way to plan, record, and review their workouts from a single account. Users can create an account, securely log in, browse an exercise library, search or filter exercises by category and muscle group, and create, use, and delete workout templates.

When users log into the app, they can immediately start to record a workout session either manually or from an existing template. Within that workout, users can add exercises from the library and log individual sets for each exercise. Each set of an exercise includes a weight and number of repetitions associated with it, entered in by the user. Once a workout session is completed, it is saved to the user's account and can be viewed later in their workout history. Over time, statistics provide insight on their overall workout activity.

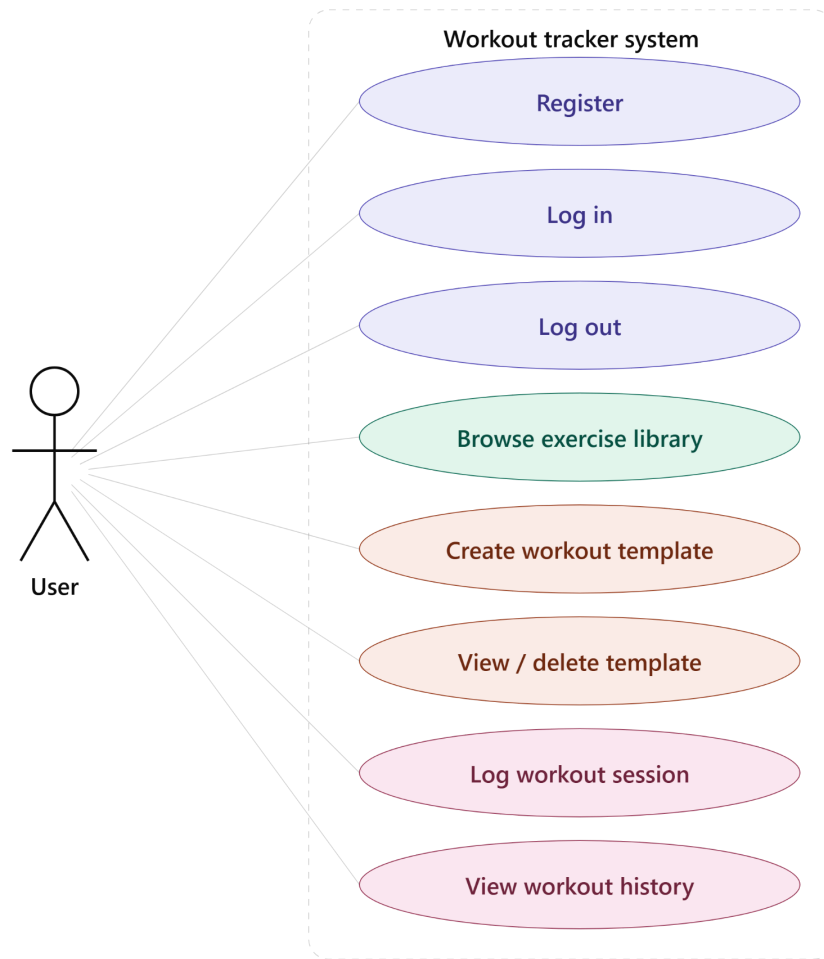
2. Functional Requirements (10 points)

1. The system shall allow a user to register an account with a username, email, and password. (high)
2. The system shall allow a registered user to log in and log out. (high)
3. The system shall hash and securely store user passwords. (medium)
4. The system shall display a library of exercises with name, category, and muscle group. (high)
5. The system shall allow a user to search/filter the exercise library. (high)
6. The system shall allow a user to create a workout template containing one or more exercises. (high)
7. The system shall allow a user to view and delete their own workout templates. (high)
8. The system shall allow a user to log a workout session, optionally based on a template. (high)
9. The system shall allow a user to record sets (weight and reps) for each exercise in a session. (high)
10. The system shall display a user's past workout sessions, ordered by date. (high)

3. Non-functional Requirements (10 points)

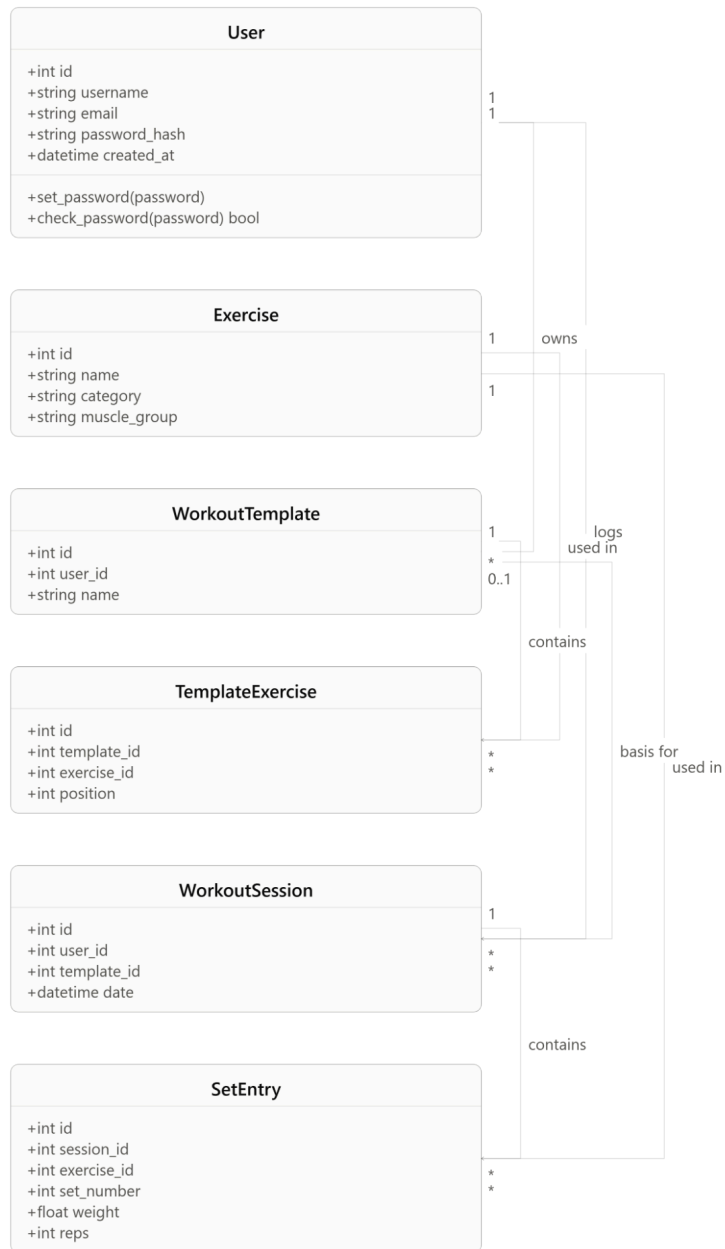
- Security: passwords must never be stored in plaintext; a user may only view/edit their own data.
- Usability: core flows (register -> log a workout) should be completable without instructions.
- Portability: the app must run in any modern browser without installation.
- Reliability: data entered during a session must persist even if the browser is closed and reopened.
- Performance: exercise library search results should return in under 1 second for the seeded dataset.

4. Use Case Diagram (10 points)

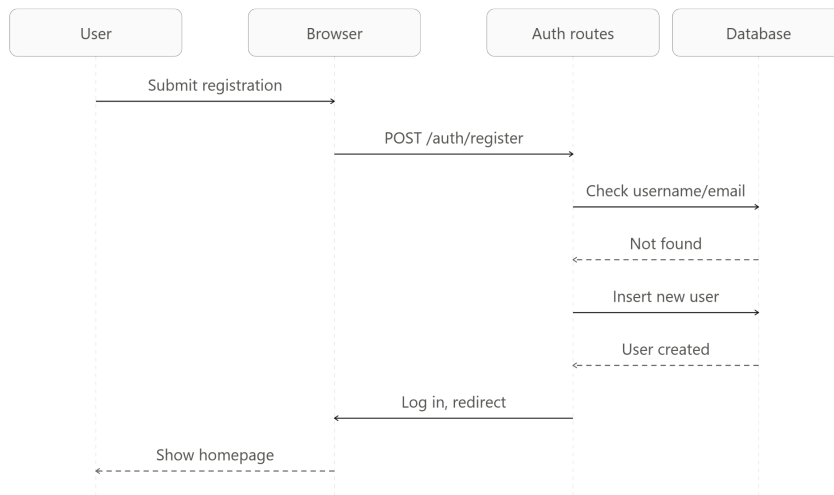


5. Class Diagram and/or Sequence Diagrams (15 points)

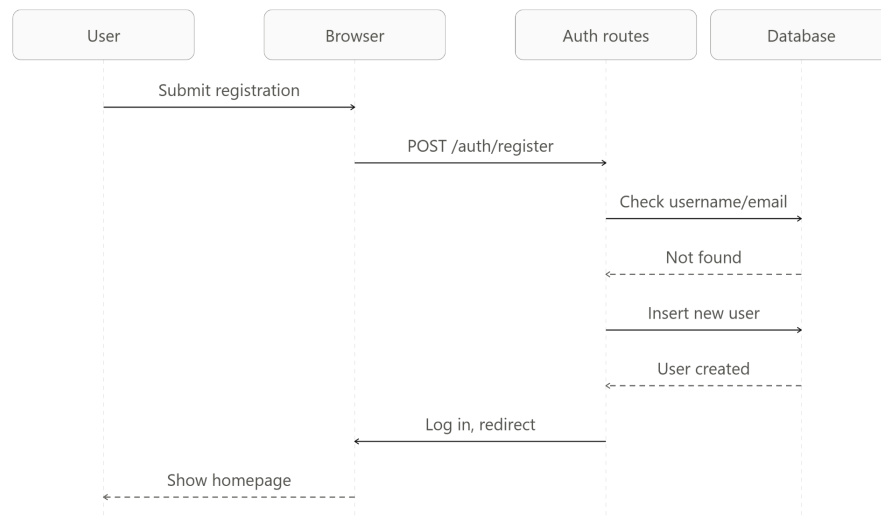
Class Diagram: Database Models



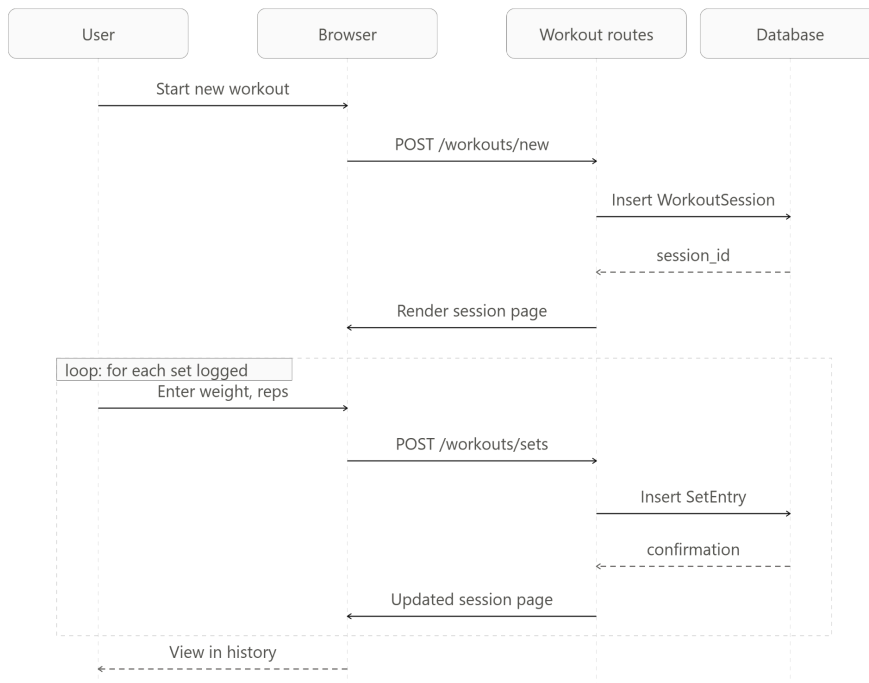
Sequence Diagram: Login and Authentication Sequence



Sequence Diagram: Workout Template Creation



Sequence Diagram: Logging a Workout



6. Operating Environment (5 points)

The software should be able to operate on any hardware platform/OS/versions as long as they are able to access a web browser and enable Javascript. It must be able to coexist with the web browser and eventually servers holding the database.

7. Assumptions and Dependencies (5 points)

Assumptions

- User understands how to use web browser UI's to a reasonable extent
 - Drop menu of options, navigation bar, etc.
- All users have a working local Python 3.10+ environment
- SQLite is sufficient for scale

Dependencies

- See requirements.txt
 - specific versions of dependencies and environment may be required
- No 3rd party APIs
- No cloud hosting/deployment dependencies